

Trig substitution — 3.3

When the integrand has a term of the form $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$, $\sqrt{x^2 - a^2}$ $a \geq 0$

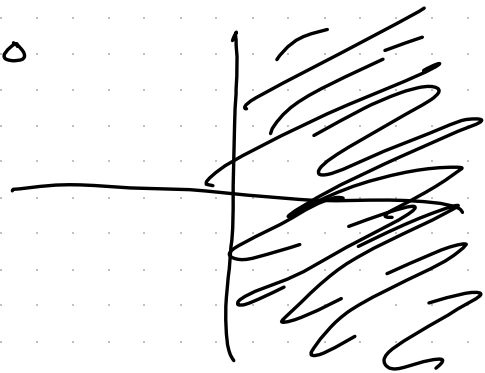
$$\begin{array}{lll} x = a \sin \theta & x = a \tan \theta & x = a \sec \theta \\ \theta = \arcsin\left(\frac{x}{a}\right) & \theta = \arctan\left(\frac{x}{a}\right) & \theta = \operatorname{arccsc}\left(\frac{x}{a}\right) \end{array}$$

$$\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2 \sin^2 \theta} = \sqrt{a^2 (1 - \sin^2 \theta)} = \sqrt{a^2 \cos^2 \theta}$$

$$= |a \cos \theta|, \quad \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \Rightarrow \cos \theta \geq 0$$

$$= a \cos \theta$$

$$\sqrt{a^2 - x^2} = a \cos \theta$$



Set $x = a \tan \theta$ ($\theta = \arctan(\frac{x}{a})$) $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$

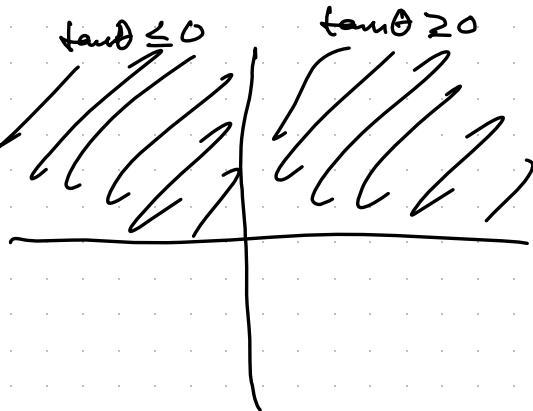
$$\begin{aligned} \sqrt{a^2 + x^2} &= \sqrt{a^2 + a^2 \tan^2 \theta} = \sqrt{a^2(1 + \tan^2 \theta)} = \sqrt{a^2 \sec^2 \theta} \\ &= |a \sec \theta| = a \sec \theta. \end{aligned}$$

Set $x = a \sec \theta$ ($\theta = \operatorname{arcsec}(\frac{x}{a})$) $\theta = [0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$

$$\sqrt{x^2 - a^2} = \sqrt{a^2 \sec^2 \theta - a^2} = \sqrt{a^2(\sec^2 \theta - 1)} = \sqrt{a^2 \tan^2 \theta}$$

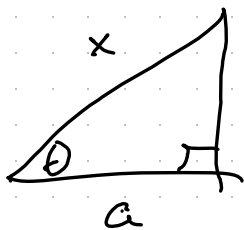
$$= |a \tan \theta| = \begin{cases} a \tan \theta & \text{if } \theta \in [0, \frac{\pi}{2}) \\ -a \tan \theta & \text{if } \theta \in (\frac{\pi}{2}, \pi] \end{cases}$$

$$= \begin{cases} a \tan \theta & \text{if } x \geq a \\ -a \tan \theta & \text{if } x \leq -a \end{cases}$$



Domain of $\sqrt{x^2 - a^2}$ is $(-\infty, -a] \cup [a, \infty)$
 $x \leq -a$ $x \geq a$

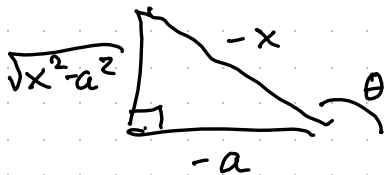
If $x \geq a$, setting $\sec \theta = \frac{x}{a}$, then the reference triangle is the usual one:



and then $\tan \theta = \frac{\sqrt{x^2 - a^2}}{a}$

$$\sqrt{x^2 - a^2} = a \tan \theta$$

If $x \leq -a$, then reference triangle is



and $\tan \theta = \frac{\sqrt{x^2 - a^2}}{-a}$

$$\sqrt{x^2 - a^2} = -a \tan \theta$$